Juhi Damley

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Computer Science junior who solves problems and builds communities

Education

Claremont McKenna College / Harvey Mudd College — Claremont, CA
Bachelor of Arts, Computer Science (cross-campus major at HMC) — August 2024 – May 2028

Coursework: Digital Electronics & Computer Engineering, Computability & Logic, Computer Systems, Databases, Stochastic Operations Research, Abstract Algebra, Linear Algebra, Statistics, Data Structures, Religion, Politics, and Global Violence

Research

Electoral Equilibrium | Researcher — May 2026 – Present
Claremont McKenna College Summer Research Program · Supervised by Prof. Gaston Espinosa

- Led an end-to-end ML pipeline modeling demographic voting-bloc shifts after political shocks, integrating a QLoRA fine-tuned Mistral-7B, CVXPY quasi-convex optimization, and Monte Carlo simulation, deployed to a live web app
- Designed a controlled "mismatch" evaluation to test whether a distilled model could learn a target reasoning capability, isolating genuine reasoning from pattern-matching and establishing that 3 frontier models across 2 labs failed the target task in 100% of 60 trials
- Diagnosed a suspected metric failure to its true root cause, distinguishing a real model limitation from a measurement artifact and preserving confidence in the evaluation pipeline

The Gould Center for Humanistic Studies | Research Assistant — October 2024 – Present

- Delivered quantitative research and 70+ data visualizations for Prof. Espinosa's national "Religion, Race, Gender, and the 2024 Election" series, presented at conferences and in a five-book series
- Built full-stack research website (TypeScript/React/Supabase)
- Managed logistics for a multi-speaker national conference

Financial Economics Institute | Research Analyst — January 2026 – May 2026

- Built a config-driven Python portfolio optimization pipeline with deterministic, reproducible runs on CMC HPC for Prof. Benjamin Gillen's Optimized Indexation project
- Identified that nested-dictionary data structures were bottlenecking the portfolio pipeline, presented a NumPy vs. pandas trade-off analysis, and drove adoption of a pandas-based redesign, cutting runtime by 20x

Leadership

Harvey Mudd College | Student Advisory Board — September 2026 – Present

- Represent student interests to faculty and staff, advise curriculum development, and build a CS community

Harvey Mudd College | Grading & Tutoring Shift Lead — August 2026 – Present

- Lead a team of 18 tutors and graders supporting 440 students across the CS department, host workshops, and facilitate data collection

Girls Who Code at the Claremont Colleges | Founder & President — March 2026 – Present

- Launched the 5C chapter of Girls Who Code from zero to a running organization: recruited and manage a 12-person officer team, develop workshops, and shipped the chapter website (gwc5c.org) with a members-only portal

Claremont McKenna College | Student Representative, Board of Trustees — September 2025 – May 2026

- Represented the student body on the Board of Trustees Advancement Committee, providing input on strategic decisions, and advising on AI in advancement

Work Experience

Claremont McKenna College | Office of Advancement Intern — October 2024 – December 2025

- Built a Gemini-based classification model that automated categorization of 4,000+ alumni records, cutting processing time 50%+
- Owned the alumni data pipeline across third-party platforms and legacy archives; built Python automation that cut manual updates in half

Projects

- **LectureTeX** | lecturetex.juhi.studio Python, FastAPI, Modal — 0→1 AI product that turns any lecture — recording, video link, or PDF — into structured, compile-ready LaTeX study notes, owning scope across web, desktop, and CLI while holding cost under $0.25 per lecture through tiered model routing
- **Prioritize** | ptz.juhi.studio TypeScript — Real-time collaborative priority-queue task manager letting teams organize and rank work by urgency with a drag-and-drop interface, calendar integration, and Chrome app
- **WakeMe** | GitHub Swift — Accessibility app for narcolepsy: monitors Apple Watch biosensor streams to detect daytime sleepiness onset and trigger real-time haptic wakefulness interventions

Skills

Languages	Python, TypeScript, Swift, Java, C++, C, SystemVerilog, R, SQL
AI / ML	PyTorch, HuggingFace Transformers, LLM Fine-Tuning & Evaluation, Prompt Engineering, Claude Code, Codex, MCP
Tools & Infra	FastAPI, Next.js, Vercel, Git, Figma, Notion, React Native, Supabase

Honors & Awards

2026-2027 Handley Leadership Fellow • National Merit Scholar • Venture Capital Program Finalist • CMC SP '25 Statistics Competition Winner • AHS Merit Scholarship ($85,000)